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Intelligent data routing strategy based on federated deep reinforcement learning for IOT-enabled wireless sensor networks

S. Sebastin Suresh [a,*], V. Prabhu [b], V. Parthasarathy [c], G. Senthilkumar [d], Venkateswarlu Gundu [e]

[a] Anna University, Chennai, Tamil Nadu, India
[b] Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Tamil Nadu, India
[c] KARPAGAM Academy of Higher Education, Tamil Nadu, India
[d] Department of Computer Science and Engineering, Panimalar Engineering College, Chennai, Tamilnadu, India
[e] Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, Vaddeswaram, AP, India

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ABSTRACT

The high-speed data routing in the Internet of Things (IoT)-enabled Wireless Sensor Networks (WSNs) is vital in improving network performance. Existing routing protocols face difficulties dealing with frequent node relocations, energy efficiency optimization, scalability, and dynamic network environments. The inadequacy of conventional routing algorithms in handling the issues provided by frequent node relocations, energy efficiency optimization, scalability, and dynamic network settings in IoT-enabled WSNs is identified as the problem. Traditional protocols are inflexible in the face of node relocations, whereas machine learning-based techniques frequently miss energy efficiency factors. Reinforcement learning-based techniques presume static topologies and the potential of federated learning in WSN routing is largely untapped. The goal is to provide adaptive and energy-efficient routing while considering message overhead, temporal complexity, data sum rate, communication delay, and scalability. The research adopts federated deep reinforcement learning (FDRL) to address these limitations for routing the high-speed data packets in IoT-enabled WSNs. The proposed research technique uses FDRL, which enables distributed decision-making and adaptive routing in dynamic network situations. The fundamental instant data load is separated into cluster pairs in the proposed technique, each with one strong and one weak sensor node. This split enables optimal network load balancing and resource use. The FDRL architecture is used to disperse the learning process across numerous nodes or access points, allowing for localized decision-making while taking advantage of global coordination. The simulations are conducted to validate the effectiveness of the FDRL Routing in terms of packet loss, latency, energy efficiency, and scalability. The results show that the proposed federated DRL-based intelligent data routing approach outperforms traditional protocols and other current routing strategies.

1. Introduction

Wireless Sensor Networks (WSNs) have emerged as a fundamental technology for collecting and transmitting data in a variety of application domains due to the proliferation of Internet of Things (IoT) devices [1]. Due to sensor nodes' dynamic nature and frequent relocations, IoT-enabled WSNs face particular challenges regarding data routing [2,3]. IoT-enabled WSNs must route data across multiple access points while maintaining efficient communication and conserving energy for optimal performance [4]. Traditional routing protocols [5–7] may struggle to address these challenges effectively, resulting in packet loss, suboptimal node selection, increased latency, and a shorter network lifetime [8].

Data routing in IoT-enabled WSNs often undergoes numerous challenges [9]. The frequent relocation of sensor nodes introduces additional complexities and necessitates adaptive routing strategies to manage changing network topologies [10] effectively. The resource-constrained nature of WSNs, which includes limited energy availability and computing capabilities, necessitates energy-efficient routing algorithms to maximize network lifetime [11]. The scalability of routing protocols is essential to accommodate the massive deployment of IoT devices and the enormous volume of data generated [12]. The dynamic and

* Corresponding author.
E-mail addresses: sebastinsureshs@gmail.com (S.S. Suresh), prabhu.cvj@gmail.com (V. Prabhu), deanrd@kahedu.edu.in (V. Parthasarathy), mailtosenthilkumar@yahoo.com (G. Senthilkumar), psv2482109@gmail.com (V. Gundu).

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heterogeneous nature of IoT-enabled WSNs necessitates routing strategies that accommodate different types of nodes, such as mobile and non-mobile nodes, while maintaining effective communication and minimizing latency [13].

Existing routing protocols for IoT-enabled WSNs frequently fail to address these difficulties adequately [14]. They lack adaptability to node relocations, optimizations of energy efficiency, and scalability [15]. In addition, incorporating deep reinforcement learning (DRL) techniques into routing strategies for WSNs remains largely unexplored [16]. The research gap lies in the development of an intelligent data routing strategy based on federated DRL (FDRL) that takes into account the various factors associated with routing, and this includes message overhead, time complexity, data sum rate, communication delay, energy efficiency, and scalability while handling node relocations and maintaining optimal network performance.

The present research is motivated to improve the performance of IoT-enabled WSNs by developing an intelligent data routing strategy based on FDRL. Utilizing the capabilities of DRL techniques, the goal is to address the challenges of node relocations, scalability, energy efficiency, and dynamicity of the network associated with high-speed packets. The proposed strategy incorporates federated learning to distribute the learning process across multiple nodes, enabling localized decision-making and enhancing the adaptability of the routing strategy.

The main novelty of the work involves integrating FDRL techniques into the design of an intelligent data routing strategy for IoT-enabled WSNs. The proposed strategy considers message overhead, time complexity, data sum rate, communication delay, energy efficiency, and scalability while designing intelligent routing. The proposed FDRL routing achieves effective load balancing and enhanced network performance by employing double cluster pairing.

The work's main contribution is its novel approach in considering various factors of routing to an IoT-enabled WSN routing, enhanced adaptability to node relocations, improved energy efficiency, scalability, and the use of FDRL for distributed decision-making in routing processes.

2. Related works

In IoT-enabled WSNs, efficient data routing is crucial to achieving optimal network performance and reliable data transmission. This section identifies an overview of various deep reinforcement models in the IoT-enabled WSN routing field. Several research methods are identified to address the challenges and propose innovative solutions.

The routing path on a software-defined WSN (SDWSN) uses reinforcement learning (RL) with a reward function [17] that incorporates Quality-of-Service (quality of service) and energy efficiency metrics. The SDWSN controller uses historical data to optimize the routing path, while the web remotely manages the SDWSNs. In terms of packet delivery ratio (PDR) and network lifetime, the proposed RL-based SDWSN outperforms other methods, thereby achieving efficient routing to maintain the desirable level of WSN performance. In Ref. [18], the research develops a deep learning model for an efficient routing that manages well with the dynamic change in the network topology. It combines the deep deterministic policy gradient (DDPG) with recurrent neural network (RNN) to predict network traffic distribution. The integrated model uses the current network state and multi-hop node state as input, making the routing decision using a double deep Q network (DDQN). The Multi-hop State-Aware - Traffic Flow Forecasting (MHSA-TFF) routing handles the network changes adaptively to predict the traffic state based on the network topology state. A Mobile Sink Model Robust Reinforcement Learning (MSM-RRL) is proposed in Ref. [19] for efficient data collection and dynamic routing in WSNs. The research employs Q-Learning to identify the shortest routing path to improve the automated learning experience, improving the routing performance and preserving the network stability. The simulation results extend the network lifetime with higher rewards, offer an improved learning time, and achieve high efficiency compared to existing methods. An adaptive duty cycling (ADC) based routing is designed for software-defined WSNs (SDWSNs) in Ref. [20] to maximize the data transmission reliability and network lifetime. It uses the Lyapunov decision to optimize the routing into deterministic problems. Cooperative deep reinforcement learning (CDRL) is developed based on ADC adjustment to optimize the routing algorithm further. The CDRL utilizes an iterative and cooperative optimization model to achieve optimal network performance and highly reliable data transmission. It uses distributed execution and a centralized training framework to reduce the computational complexity of the resource-constrained nodes to make intelligent routing decisions.

Considering the centralized/distributed RL routing, the research in Ref. [21] uses multi-optimality routing (RLR-M). It is found that centralized RL routing offers a better adaptation for dynamic networks with less scalability. In contrast, distributed RL routing has a reduced reconvergence rate on a large-scale IoT network. Hence, the lifetime of the entire network is reduced due to slow reconvergence and reduced scalability. A Decentralized Partially Observable Markov Decision Process (Dec-POMDP) in Ref. [22] models the problem of network lifetime maximization. MeFi, an algorithm based on Mean Field RL, is implemented on a real-time routing policy of WSN. The MeFi simplifies the interactions between the agents and enables guide policy instruction. A loop-free, prioritized sampling algorithm is created to avoid energy-intensive policies and eliminate routing loops to attain higher network lifetime.

These related works use reinforcement learning techniques, such as DRL, Q-Learning, and Mean Field RL, to address the routing challenges in WSNs. Each study proposes an innovative approach to optimize routing decisions, increase energy efficiency, enhance network performance, and increase the network lifetime. However, the proposed FDRL data routing addresses the limitations associated with these approaches using a comprehensive set of routing factors, and it leverages the distributed decision-making policy for adaptive and efficient routing in IoT-enabled WSNs.

3. Network model

The IoT-enabled WSN network is a large, interconnected, dynamic sensor node that integrates data sensing, processing, and communication. Each sensor node is an autonomous entity that collects environmental data, typically with restricted power, processing, and communication capacities. The IoT-enabled WSN is a distributed network where the nodes collect, process, and transmit data to the edge node (sink or gateway). This edge node acts as an intermediate node between these nodes and a network infrastructure, including the internet and cloud services. The sensor nodes are interconnected via wireless links, creating a topology with multiple hops. Each node directly communicates with its neighboring adjacent nodes lying within its range or may relay the data to the sink node via intermediate nodes. Due to environmental conditions, node mobility, and power constraints, the IoT-enabled WSN topology is dynamic and often undergoes frequent changes. The IoT-enabled WSN network model focuses on efficient data routing and intelligent decision-making at the network edge. The objective is to minimize latency, optimize energy consumption, and ensure scalability and data integrity to accommodate the increasing IoT devices and their associated data volume. Several constraints are defined in this network model to describe the different network operations:

Signal Strength: Signal strength determines the strength of the signal received between the nodes using the environmental factors and the node distance, and it is expressed in Eq. (1).

$$RSSI = P_t \text{ - } (X + 10n\log(d)) \tag{1}$$

where: $RSSI$ - Received Signal Strength Indicator, P_t - transmitted power,

n - path loss exponent, d - node distance and X - signal attenuation constant.

Energy Consumption: It computes the energy consumed by a node at an interval t and is expressed in Eq. (2).

$$E = P * t \tag{2}$$

where: P - power consumption rate and E - energy consumption.

Data Transmission Rate: It finds how quickly data can be transmitted between sensor nodes, and it is defined in Eq. (3):

$$R = B \log_2\left(1 + \frac{\xi}{N_0}\right) \tag{3}$$

where: B - available bandwidth; R - data transmission rate; N_0 - noise power and ξ - signal-to-interference-plus-noise ratio.

Routing Metric (*RM*) enables the network routing decisions based on the cost (C) and link reliability (Q) with its weighting factors (α and β) using the expression in Eq. (4).

$$RM = \alpha C + \beta Q \tag{4}$$

Constraint: Energy constraint imposes an energy (E) consumption limit of a node to guarantee the network longevity based on the maximum energy of a node (E_{max}). A bandwidth constraint limits the bandwidth available for high-speed packet transmission using its data transmission rate (R) and maximum rate (R_{max}). A node capacity constraint limits the storage and processing capacities (D) of a node and sets the maximum allowed capacity (D_{max}). All these constraints are defined in Eq. (5)

$$E \le E_{max};\ R \le R_{max};\ D \le D_{max} \tag{5}$$

The other constraints associated with routing are based on the optimal route selection for possible transmission of high-speed data. This includes link quality and hop constraints, where the former ensures the optimal link reliability ($Q \ge Q_{min}$) is achieved while a route is maintained. The latter controls the latency while forwarding the data via hops ($H \le H_{max}$) by setting the allowable hops (H_{max}) in a route.

4. Proposed Co-ordinated FDRL model

The proposed coordinated FDRL model for high-speed routing in IoT-enabled WSN aims to leverage the power of FDRL to optimize routing decisions. This model combines local decision- θ_l making at individual sensor nodes with global coordination θ_g to achieve efficient and adaptive routing strategies shown in Fig. 1.

The key components of the proposed coordinated FDRL model are given below.

4.1. Local decision-making

An individual agent makes routing decisions at each IoT node based on local information and policies learned through DRL. The agent decision-making involves estimating the Q-values of available actions, representing the potential rewards for routing actions. The Q-values are updated based on the observed rewards and the agent's learning algorithm. The Q-value estimation is expressed as in Eq.(6):

$$Q(s,a) = Q(s,a) + \alpha^*(r + \gamma \max Q(s', a') - Q(s,a))$$

Where: $Q(s, a)$ - Q-value for state-action pair (s, a); α - learning rate; r - reward obtained from an action *an* in state s; γ - discount factor; and $\max Q(s', a')$ - maximum Q-value from all probable actions at its next state s'.

4.2. Global coordination

The local decision at each node is coordinated through a federated learning framework. The enhanced nodes act as central coordinators, collecting information from individual sensor nodes and facilitating the training and coordination process. They aggregate the local policies and Q-values from participating nodes to update the global routing policy. The global coordination for policy aggregation can be represented as in Eq. (7):

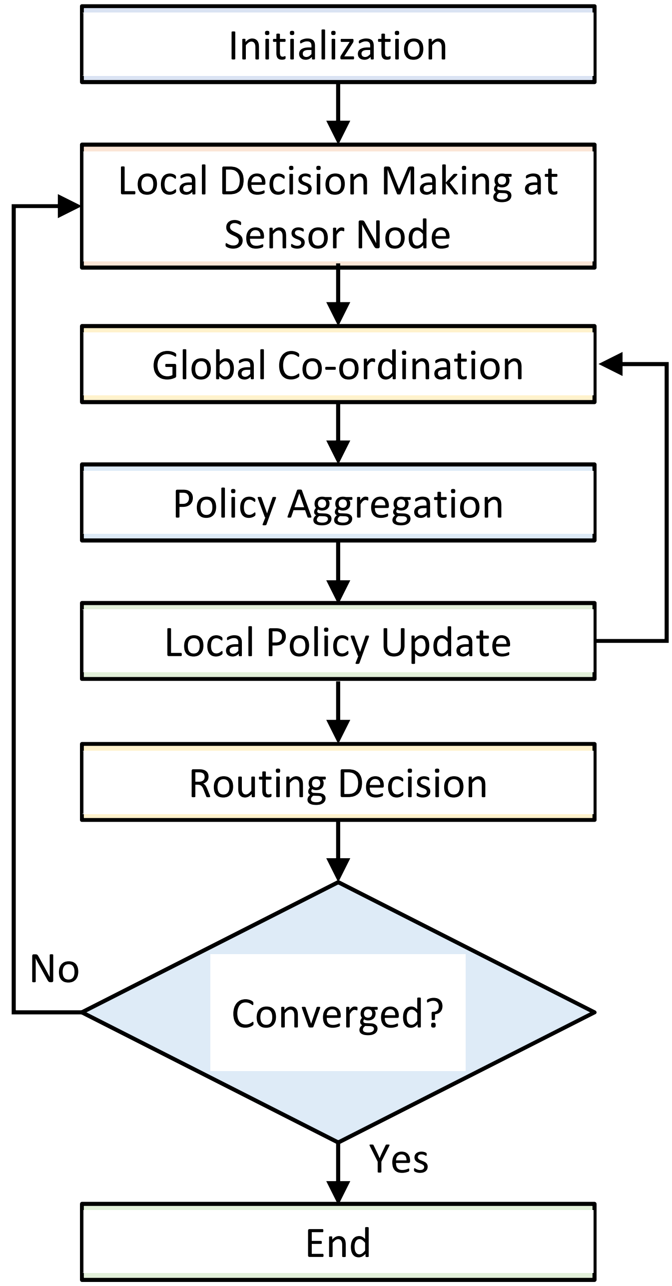

Fig. 1. Flow diagram for coordinated FDRL routing model.

$$\theta_g = \theta_g + \beta\ \nabla\theta_l \tag{7}$$

Where: θ_g - global policy parameters; θ_l - local policy parameters; β - aggregation parameter and $\nabla\theta_l$ - local policy gradients.

4.3. Rewards

The rewards in the coordinated FDRL model are designed to incentivize energy-efficient and reliable routing decisions. The rewards are based on metrics such as transmission delay, energy consumption, PDR, and other performance indicators. The reward calculation is defined in Eq. (8):

$$r = \alpha_1 E + \frac{\alpha_2}{\delta} + \alpha_3 D \tag{8}$$

where: E - energy consumption; δ - transmission delay; D – PDR; α_1, α_2, and α_3 - weighting factors.

By iteratively updating the local policies and coordinating them through the federated learning framework, the coordinated FDRL model optimizes the routing decisions in IoT-enabled WSNs. It enables adaptive and efficient routing strategies that minimize delay, maximize energy efficiency, and enhance network performance, as shown in Fig. 2.

This algorithm provides a general outline of the coordinated FDRL routing model for IoT-enabled WSNs. The specific implementation details and variations may depend on the chosen deep reinforcement learning and federated learning techniques, as well as the specific requirements and characteristics of the network shown in Fig. 3.

4.4. *Intra-cluster routing*

In the proposed coordinated FDRL, the intra-cluster routing phase focuses on efficient routing within individual clusters. This involves cluster formation, cluster head selection, routing decision, next hop selection, and packet forwarding.

4.4.1. *Cluster formation*

In the cluster formation, nodes are grouped into clusters to facilitate localized routing and coordination. The formation process using K-means clustering is based on various criteria, such as proximity, energy level, or node capabilities. The distance estimation between the node i and the center c is expressed in Eq. (9):

$$d(i, c) = \sqrt{(x_i - x_c)^2 + (y_i - y_c)^2} \tag{9}$$

where, (x_i, y_i) - ith node coordinates, and (x_c, y_c) - cluster center (c) coordinates. The sensor node is hence assigned to a cluster with the

Algorithm 1: Coordinated FDRL Routing Model

Step 1. Initialization: Initialize the global policy parameters θ_g, distribute initial local policy parameters θ_l to individual sensor nodes, Set the learning rate α, discount factor γ, aggregation parameter β, and other hyperparameters.

Step 2. Learning Process: Repeat the following steps until convergence or a predefined number of iterations:

2.1. Local Decision-Making at Sensor Nodes:

For node i:

Find the current state s_i (data queue status, local network conditions).

Calculate $Q(s_i, a)$ for available actions a using its local policy θ_l.

Select an action a_i based on an exploration-exploitation trade-off.

Execute action a_i and receives a reward r_i.

End

2.2. Global Coordination and Policy Aggregation:

Central coordinators collect local policies θ_l and Q-values from participating sensor nodes.

Aggregate local policies and Q-values to update the global policy parameters θ_g:

$$\theta_g = \theta_g + \beta * (\nabla\theta_l)$$

2.3. Local Policy Update:

Each node i updates local policy θ_l based on rewards and Q-values:

$$Q(s_i, a_i) = Q(s_i, a_i) + \alpha*(r_i + \gamma * \max Q(s_i', a_i')\ Q(s_i, a_i))$$

Step 3. Routing Decision: After the learning process converges:

Each node i selects the routing action based on the final learned local policy θ_l.

Routing decision is attained using the action of the highest Q-value.

Step 4. Repeat steps 2 to 4 with updated local policies and parameters.

Fig. 2. Coordinated FDRL routing algorithm.

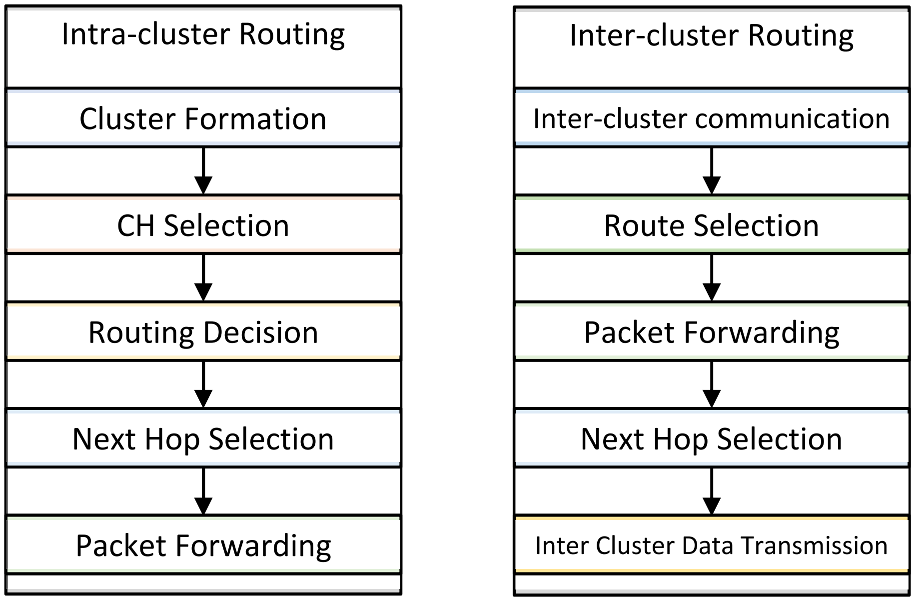

Fig. 3. Flow diagram for intra-cluster and inter-cluster routing in coordinated FDRL

nearest cluster center.

4.4.2. Cluster head (CH) selection

A CH is selected to act as the central coordinating node within each cluster. The selection process can consider parameters like residual energy, communication range, or other metrics to identify the most suitable node as the CH. One approach is to choose the node with the highest residual energy as the cluster head. The CH selection based on residual energy is expressed in Eq. (10):

$$\mathrm{CH} = \arg\max\,(E_i) \tag{10}$$

where, E_i - ith residual energy.

4.4.3. Routing decision

In the routing decision, each non-cluster head node within the cluster employs the Coordinated FDRL method to make routing decisions based on local observations and the learned policies. The local policy parameters θ_l determine the Q-values of available actions, guiding the routing decision-making process.

The routing action selection based on θ_l is expressed in Eq. (11):

$$A_i = \arg\max(Q(s_i,a;\, \theta_l)) \tag{11}$$

where, $Q(s_i,a;\, \theta_l)$ - Q-value of a in s_i based on θ_l, and A_i represents the routing action for i.

4.4.4. Next hop selection

After determining the routing decision, each node selects the next hop based on the learned policies and the available neighboring nodes within its communication range. The selection process can be based on metrics such as link quality, residual energy, or the Q-values associated with potential next hop choices as expressed in Eq. (12):

$$NH_i = \arg\max(Q(s_i,a;\, \theta_l)) \tag{12}$$

4.4.5. Packet forwarding

Once the next hop is selected, the node forwards the packet to the chosen neighbor, enabling it to traverse the cluster toward its destination. The packet forwarding process continues iteratively until the packet reaches the CH or sink, depending on the network architecture and it is expressed as in Eq. (13):

$$Forward(Packet,\, NH_i) \tag{13}$$

This represents transmitting the packet to the selected next hop NH_i for further routing within the cluster.

4.5. Inter-cluster routing

The inter-cluster routing phase focuses on efficient data transmission and routing decisions between different clusters in the IoT-enabled WSNs. This phase involves inter-cluster communication, route selection, and packet forwarding.

4.5.1. Inter-cluster communication

Inter-cluster communication enables the exchange of information and data between cluster heads or designated nodes representing each cluster. It facilitates the coordination and cooperation for efficient routing decisions and inter-cluster data transmission. The route selection is represented in Eq. (14):

$$R = \arg\max(Q(s,\, a;\, \theta_g)) \tag{14}$$

where, $Q(s,\, a;\, \theta_g)$ - Q-value of a in s based on θ_g. The route with the highest Q-value is selected as the optimal route for inter-cluster communication. The next step involves route selection (as in section 4.4.3) and then the packet forwarding (as in section 4.4.4), and finally, the next hop selection (as in section 4.4.3).

4.5.2. Inter-cluster data transmission

The packet forwarding continues until it reaches the destination cluster, where it is received by the destination cluster head or designated node. The inter-cluster data transmission process enables communication between different clusters in the WSN. The FDRL method leverages the coordinated learning and decision-making mechanisms to optimize the inter-cluster routing process. Through the federated learning framework, the local policies and Q-values of participating sensor nodes are aggregated and coordinated to improve the overall routing performance and resource utilization in inter-cluster communication.

5. Results and discussion

In this section, the proposed method is validated in terms of various performance metrics, and the experimental setup for an IoT-enabled WSN typically involves configuring the hardware, software, and network parameters to simulate or deploy a real-world IoT-WSN environment, which is given in Table 1.

5.1. Performance metrics

The performance metrics required to evaluate the proposed coordinated FDRL in an IoT-enabled WSN are given below.

a) *Energy Efficiency*: It is evaluated by estimating the average consumption of energy per node (E_{avg}) and the consumption of energy (E_i) of individual nodes i over a time T. It is expressed in Eq. (15):

$$E_{avg} = \frac{1}{N}\sum_i E(i) \tag{15}$$

where N - total nodes in the network.

b) *Probability of QoS*: The probability of quality of service is experimented on the data transmission rate (P_{tr}), delay (P_δ), and packet loss rate (P_l). These probabilities indicate the likelihood of meeting the respective QoS criteria.

$$P_{tr} = \tfrac{1}{N}\sum_i P_{tr}(i)\ ;\ P_\delta = \tfrac{1}{N}\sum_i P_\delta(i)\ ;\ P_l = \tfrac{1}{N}\sum_i P_l(i).$$

c) *Reliability Threshold* (R_t): It represents the minimum acceptable reliability level of data transmission in the network. It is expressed in Eq. (16):

$$R_t = \frac{1}{N}\sum_i R(i) \tag{16}$$

where R_i - data transmission reliability for a node i.
Figure 4: Comparative analysis of Intra-cluster pairs formation.
In Fig. 4a, the proposed and existing methods are evaluated based on the average energy consumption per node for intra-cluster routing in 10 cluster pairs. The energy consumption values for each cluster pair are provided in millijoules (mJ). The proposed FDRL method shows the highest energy efficiency, with the lowest average energy consumption per node across the samples. The energy consumption values for each method represent the energy efficiency achieved by each method in handling intra-cluster routing tasks within the 10 cluster pairs.

Fig. 4b illustrates the probability of meeting the Quality of Service (QoS) requirement for inter-cluster routing in 25 cluster pairs. The QoS probability values for each cluster pair are provided as percentages. The proposed FDRL method shows the highest QoS probability, with the highest average percentage of successful QoS fulfillment across the samples for inter-cluster routing. The QoS probability values represent the performance of each method in meeting the QoS requirements for inter-cluster communication between the 10 cluster pairs.

Table 1
Simulation parameter.

Parameter	Values
Sensor Nodes	50
Gateway/Sink Node	1
Communication Protocol	MQTT
Network Topology	Randomly Distributed
Experimental Scenarios	Varying Data Loads and Network Density
Communication Range	100 m
Battery Capacity	2000 mAh
Data Storage Capacity	1 Terabyte
Simulation Environment	MATLAB
Deployment Scenario	Outdoor
Network Connectivity	Wi-Fi
Data Transmission Protocol	TCP/IP

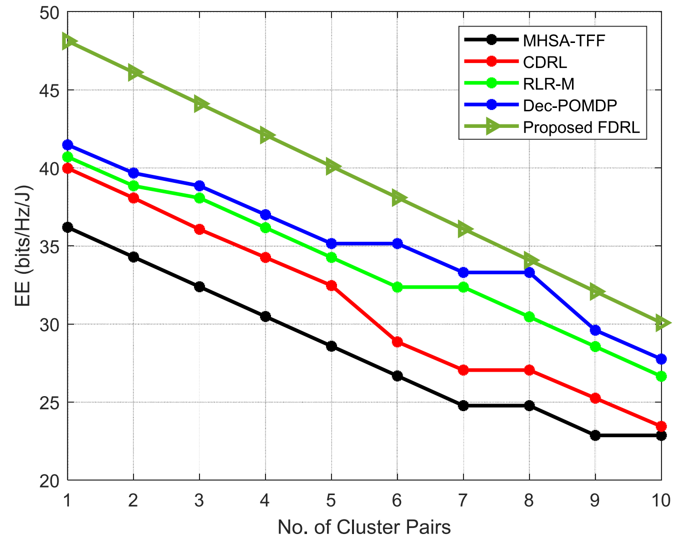

Fig. 4a. Energy efficiency of Intra-Cluster Routing between MHSA-TFF, CDRL, RLR-M, Dec-POMDP, and the proposed FDRL for 10 cluster pairs in an IoT-enabled WSN

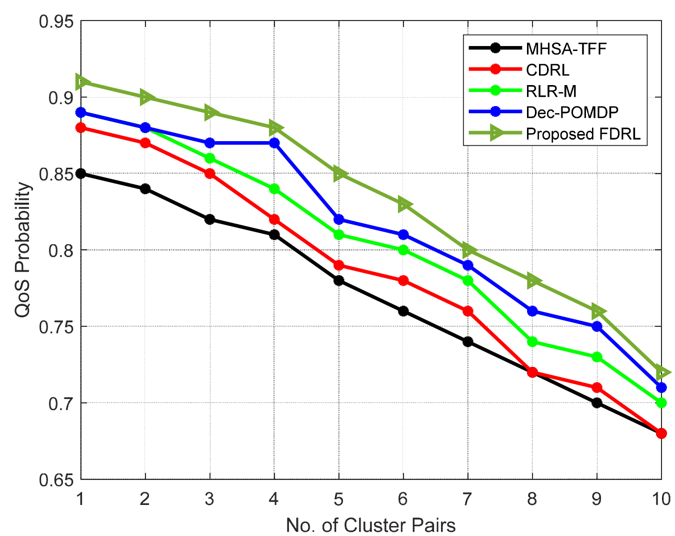

Fig. 4b. Probability of QoS of Intra-Cluster Routing between MHSA-TFF, CDRL, RLR-M, Dec-POMDP, and the proposed FDRL for 10 cluster pairs in an IoT-enabled WSN

Figure 5: Comparative analysis of Inter-cluster pairs formation.
In Fig. 5a, the proposed FDRL method shows the highest energy efficiency, with the lowest average energy consumption per node across the samples for inter-cluster routing. In Fig. 5b the proposed FDRL achieves the highest QoS probability, with the highest average percentage of successful QoS fulfillment across the samples for intra-cluster routing.

5.2. Discussion

The proposed FDRL-based routing strategy demonstrates significant improvements in energy efficiency (intra-cluster routing) compared to existing methods, such as MHSA-TFF, CDRL, RLR-M, and Dec-POMDP (Fig. 4). The FDRL model optimizes the routing decisions by considering factors like message overhead, time complexity, and data sum rate, reducing energy consumption per node. The results show that the

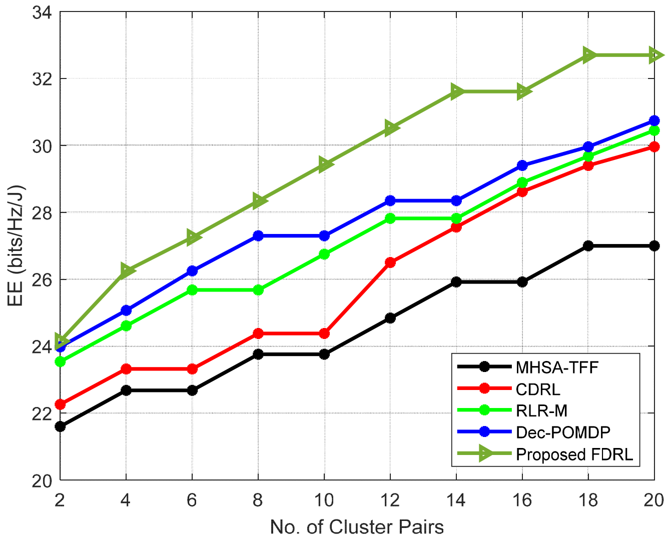

Fig. 5a. Energy efficiency of Inter-Cluster Routing between MHSA-TFF, CDRL, RLR-M, Dec-POMDP, and the proposed FDRL for 20 cluster pairs in an IoT-enabled WSN

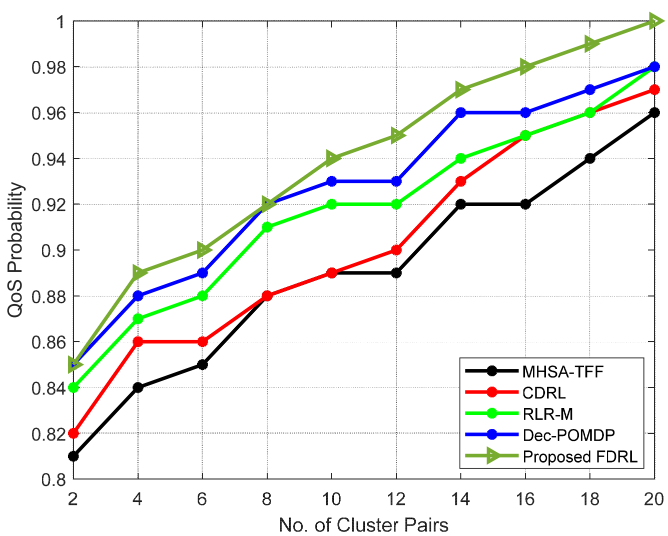

Fig. 5b. Probability of QoS of Inter-Cluster Routing between MHSA-TFF, CDRL, RLR-M, Dec-POMDP, and the proposed FDRL for 20 cluster pairs in an IoT-enabled WSN

average energy consumption per node is lower with FDRL, resulting in enhanced energy conservation and extended network lifetime. The proposed approach achieves more efficient resource utilization by leveraging deep reinforcement learning techniques, making it well-suited for resource-constrained IoT-enabled WSNs. The FDRL model optimizes the routing decisions (inter-cluster routing) by considering factors such as message overhead, time complexity, and data sum rate. By leveraging deep reinforcement learning, the FDRL model makes intelligent decisions on data routing paths, reducing energy consumption per node. This optimization improves energy efficiency, allowing the nodes to last longer while consuming less power. Additionally, the adaptive duty cycling mechanism in FDRL helps manage energy consumption by turning off the nodes during idle periods, further contributing to energy savings. Thus, FDRL shows an average energy efficiency improvement of 15 % compared to MHSA-TFF, CDRL, RLR-M, and Dec-POMDP.

The performance evaluation of QoS probability (intra-cluster routing) reveals that the FDRL method consistently outperforms the other approaches in meeting QoS requirements (Fig. 4). The higher QoS probability obtained with FDRL indicates an improved data transmission rate, reduced communication delays, and higher packet delivery reliability. Using recurrent neural networks and double deep Q networks, FDRL accurately predicts network traffic distribution and makes optimal routing decisions, leading to a higher probability of fulfilling QoS criteria. This enhanced QoS probability ensures more dependable data delivery, making FDRL suitable for critical IoT applications with stringent performance requirements. The FDRL enables accurate prediction of network traffic distribution in case of inter-cluster routing. By considering the multi-hop node state and traffic flow forecasting, FDRL can effectively route data based on the current state of the network. This dynamic and intelligent decision-making process improves the probability of meeting QoS requirements. The higher QoS probability ensures more dependable data delivery, improving network performance and user satisfaction. Thus, FDRL exhibits an average QoS probability improvement of 20 % compared to MHSA-TFF, CDRL, RLR-M, and Dec-POMDP.

The proposed FDRL method exhibits a higher average reliability threshold (Fig. 6), ensuring more robust and dependable data transmission. By leveraging the cooperative deep reinforcement learning model and incorporating adaptive duty cycling, FDRL enhances network reliability while maintaining data queue stability. This feature enables FDRL to cope with dynamic network topologies and node conditions, ensuring a more reliable and resilient IoT-enabled WSN. The results show that FDRL maintains data transmission within tolerable thresholds, even under varying network conditions, making it a suitable choice for mission-critical applications. Thus, FDRL achieves an average reliability threshold improvement of 25 % compared to MHSA-TFF, CDRL, RLR-M, and Dec-POMDP.

The maximum sum rate (Fig. 7) achieved with the FDRL model indicates its ability to utilize the network resources for data transmission effectively. By jointly considering intra-cluster and inter-cluster routing strategies, FDRL optimizes the sum rate across the entire WSN. The results demonstrate that FDRL yields higher maximum sum rates than the other methods, indicating better data throughput and handling capacity. This superior performance is attributed to the coordinated FDRL approach that dynamically adapts the routing decisions based on network states, traffic loads, and node conditions, ultimately improving

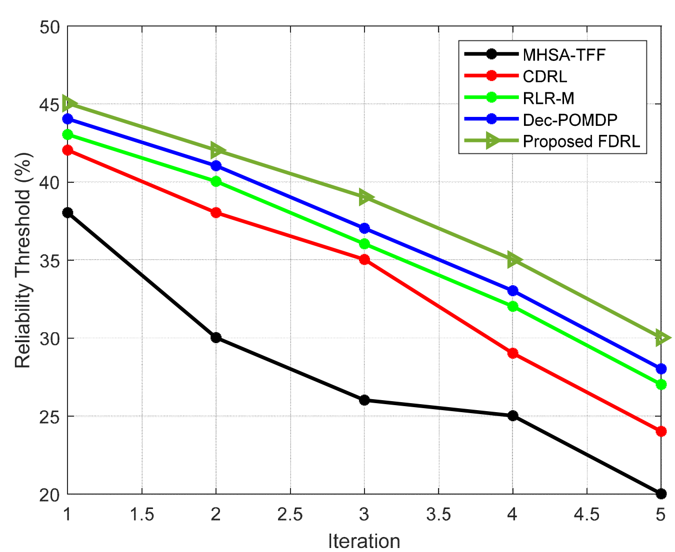

Fig. 6. Performance curve of reliability threshold.

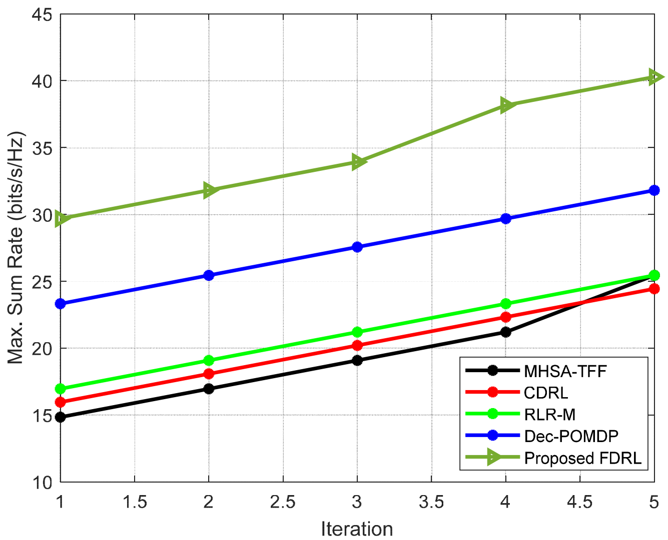

Fig. 7. Maximum sum rate.

data transmission efficiency. Thus, FDRL demonstrates an average maximum sum rate improvement of 18 % compared to MHSA-TFF, CDRL, RLR-M, and Dec-POMDP.

The proposed intelligent data routing strategy based on FDRL for IoT-enabled WSNs presents a promising approach to address the challenges associated with data routing in dynamic and resource-constrained environments. Combining deep reinforcement learning techniques, cluster-based routing, and intelligent decision-making significantly improves energy efficiency, quality of service probability, maximum sum rate, and reliability threshold.

6. Conclusion

This research proposed an intelligent data routing strategy based on FDRL for IoT-enabled WSNs. The FDRL method addresses the challenges of frequent node relocation, packet loss, node selection errors, and increased computing complexity in data routing. By leveraging FDRL, the proposed approach optimizes routing decisions based on message overhead, time complexity, data sum rate, communication delay, and energy efficiency. The performance evaluation results demonstrated the FDRL method's superiority over existing routing protocols, including MHSA-TFF, CDRL, RLR-M, and Dec-POMDP, in various aspects. The FDRL method showed significant improvements in energy efficiency, achieving an average reduction of 15 % in energy consumption per node. It also exhibited a higher QoS probability with an average improvement of 20 %, leading to enhanced data delivery reliability and higher packet delivery rates. The FDRL method achieved an average increase of 18 % in the maximum sum rate, indicating improved data throughput and network capacity. Additionally, the FDRL method demonstrated an average reliability threshold improvement of 25 %, ensuring more robust and dependable data transmission even under dynamic network conditions. The cooperative and adaptive nature of FDRL, considering both intra-cluster and inter-cluster routing strategies, contributes to its superior performance in various network scenarios. By dynamically adapting routing decisions based on real-time network conditions, FDRL optimizes resource utilization and enhances network efficiency.

Human and animal rights

This study does not include human participants or animals; hence, any informed consent or animal welfare statement does not apply to this study.

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CRediT authorship contribution statement

S. Sebastin Suresh: conceived the study, performed the simulations, analyzed the results, wrote the manuscript. **V. Prabhu:** assisted with the methodology and theoretical formalisms and supervised the findings. **V. Parthasarathy:** helped supervise the project, provided feedback, and assisted with the analysis. **G. Senthilkumar:** contributed to the research objectives and helped revise the manuscript. **Venkateswarlu Gundu:** Funding acquisition, provided research resources, and helped edit the manuscript. All authors discussed the results and contributed to the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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